SAIRAM KRISHNAN

Engineering Leader · Scaling a 100M-User AI Knowledge Graph · Wharton MBA
Seattle, WA · Open to relocation · 281-513-1263 · sairambkrishnan@gmail.com · U.S. Citizen
linkedin.com/in/sairambkrishnan · github.com/SaiWebApps · orcid.org/0009-0004-2626-1273

PROFILE

Engineering leader with 11 years of experience, rising from junior software engineer at Microsoft to senior software engineer at SIG and ultimately, engineering lead in Amazon and Apple. Currently a Principal Software Engineer at Apple, leading an 8-engineer team and scaling a media knowledge graph for 100M+ users. Personally ships production agentic AI, including an LLM ops assistant the team uses daily. Earlier at Amazon, led 8 engineers and built the Alexa Core Engine, a platform that saved $40M a year and lifted model accuracy 40%. Also achieved Wharton MBA in May 2026 and teaches AI strategy to C-suite leaders on the side through Praxtera AI Institute.

Targeting a Senior Software Engineering Manager role in AI/ML.

EXPERIENCE

Apple | Principal Software Engineer, Knowledge Graph Platform — Feb 2021 – Present

Apple Media Products · Seattle, WA · Engineering lead for an eight-engineer team across Seattle, London, and partner sites in Europe and Latin America, reporting to the engineering director

- Led architecture and cross-functional delivery of Apple's media knowledge graph, the canonical model unifying 15+ sources to power search, recommendations, and discovery for over 100 million users.
- Mentored engineers to promotion, shaped performance reviews, and drove hiring; set the design-review and CI/CD standards that lifted team velocity 30% and cut production incidents 45%.
- Built and shipped production agentic AI: an LLM ops assistant for debugging logs, traces, and metrics over MCP in natural language, and a skill platform that turned pipeline work into plain-English workflows for daily, on-call use.
- Led the four-office global launch of Apple Music Credits (Cupertino, Seattle, London, Singapore), coordinating a 15-engineer cross-functional program to deliver real-time credits curation for millions of tracks.
- Drove the platform from batch to real-time streaming, re-architecting ingestion, similarity mapping, and graph generation from Spark onto an event-driven Flink, Kafka, and Cassandra stack.
- Designed the ML entity-resolution system pairing similarity mapping with clustering to deduplicate entities across sources, lifting data quality 40% while cutting downstream cost.
- Owned the graph query services and the client library for partner teams to build on, and set the shared Protobuf contracts adopted across the platform.

Praxtera AI Institute | AI/ML Technical Leader, Executive Education — Sep 2025 – Present

Philadelphia-based AI executive-education firm · concurrent with the Apple role

- Taught AI strategy to C-suite leaders at HP, Morgan Lewis, PBS, and the Philadelphia Eagles, working from an executive curriculum built on original Wharton research to help executives evaluate, govern, and deploy AI.
- Keynoted the Philadelphia Eagles' 12th annual Eagles Care Summit at Lincoln Financial Field in 2026, delivering "AI for Impact" to more than 300 community and business leaders.

Amazon | Senior Software Engineer, Alexa — Apr 2018 – Feb 2021

Seattle, WA · Promoted from SDE II (L5) to Senior SDE (L6); led an eight-engineer team on the Alexa Core Engine

- Designed a GraphQL data-aggregation and caching platform for the Alexa Engine whose annotation-driven rule engine batched and clustered results to the right downstream services, saving $40M a year and lifting model accuracy 40%.
- Mentored half the team from entry level to promotion, earning selection for Amazon's Mentorship and Leadership Development Program.
- Collapsed partner onboarding from two weeks to 15 minutes with a self-service rapid-deployment platform.
- Owned operational excellence, driving the weekly issue count from 40 to 10 and re-engineering the new-user registration flow to cut peak-holiday customer issues tenfold.
- Led capacity planning for Prime Day and Black Friday, forecasting hardware demand and winning leadership's commitment to expand the fleet ahead of peak.

Susquehanna International Group (SIG) | Senior Financial Data Engineer Nov 2016 – Mar 2018

Philadelphia, PA · Promoted from Financial Data Engineer

- Led real-time FINRA compliance surveillance across the firm's market-making transactions in C++, Python/Pandas, and Django, flagging violations at the moment of execution.
- Designed and shipped a real-time hardware asset-tracking system that automated capacity scaling and ended chronic inter-team coordination failures.

Microsoft | Software Development Engineer, Windows OS Group Mar 2015 – Oct 2016

Redmond, WA

- Cut Windows 10 Mobile Mail power consumption 4x against the Windows Phone 8 baseline, and set the app's power targets.
- Drove product decisions with telemetry: real-time dashboards, weekly guidance to leadership, and the data-derived default for the auto-sync interval.

EDUCATION

The Wharton School, University of Pennsylvania | Executive MBA, WEMBA Class of 2026 May 2026

Palmer Scholar (top 5% of class) · Benjamin Franklin Award (peer-voted) · Director's List, all six terms · First Year Honors

Majors (5): AI in Business · Strategic Management · Operations, Information & Decision-Making · Entrepreneurship & Innovation · Business Analytics

Carnegie Mellon University | M.S. in Electrical & Computer Engineering, GPA 4.0/4.0 Dec 2014

Carnegie Mellon University | B.S. in Electrical & Computer Engineering; Minor in Business Administration May 2014

PUBLICATIONS

- Krishnan, S., & Thokala, G. B. (2026). *Who Trains the Next Senior? Managing Engineering Teams When AI Makes Code Cheap*. Working paper.
- Krishnan, S., Hepp, A., & Gandhi, S. (2026). *A Multi-Layer Framework for Evaluating the Return on Investment of AI Projects*. SSRN. doi:10.2139/ssrn.6732598
- Krishnan, S., & Ting, D. (2026). *Drone-Megaddon 2026: Co-Intelligent Swarm Control with Language and Vision Models*. engrXiv. doi:10.31224/6987
- Krishnan, S., & Ting, D. (2014). *Drone-Megaddon 2014: A Real-Time Strategy Interface for Managing Drone Swarms*. engrXiv (archived 2026). doi:10.31224/6924

INDEPENDENT R&D

Self-directed R&D: production-grade systems built and shipped solo, from architecture to deployment.

- **DependIQ:** The verification layer agentic coding still lacks. Maps the blast radius of any dependency change across a repo graph, then generates targeted, reviewable fixes. Built on FastAPI, Neo4j, and Postgres.
- **Crucible:** Adversarial QA for AI agents, open-sourced. Rival agents attack a problem from opposing angles behind an evidence-gated review, and hooks kill unsafe tool calls before execution. A Claude Code toolkit.
- **Persuaider:** A sparring partner for high-stakes negotiations. In-character personas push back in real conversation, then an LLM scores the full transcript and surfaces the moves that land. Built on Next.js, with Claude, GPT, and Gemini behind a failover chain.
- **Ondoway:** Personalized audio tours, composed on the fly. A Neo4j knowledge graph (7 entity, 11 relationship types) models travelers, places, and stories; an LLM and text-to-speech narrate each trip as travelers move. FastAPI backend, 177 tests.

SKILLS

Leadership: Engineering team leadership · Cross-functional and multi-site delivery · Mentorship and talent development · Hiring · Roadmap ownership · Executive communication and public speaking

Applied & Agentic AI: LLM application development (Claude API) · Agentic systems and MCP · Multi-provider LLM orchestration · AI-assisted engineering operations · AI strategy and ROI evaluation

Data & ML Platforms: Knowledge graphs · Entity resolution · Flink · Kafka · Spark · Cassandra · Neo4j · GraphQL · Protobuf/Avro · AWS · Distributed systems at 100M-user scale

Programming Languages: Java · Scala · Python · Go · C++ · TypeScript/JavaScript

Spoken Languages: English · Tamil · Spanish · French · Italian · Chinese